

Your grade: 100%

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Next item →

1. What quantity should a programmer maximize in order to optimize a program's performance?

1 / 1 point

☐ Latency

☐ Bandwidth

☒ Cache hits

☐ Cache evictions

Nice work

Correct. A programmer can design a program to maximize cache hits, thereby increasing the processor's efficiency and the program's performance.
2. Which of the following approaches is probably the least effective in increasing a program's memory efficiency?

1 / 1 point

☐ When working with multidimensional arrays, ensuring the program is iterating through them in the same order (row-major or column-major) as the order in which they're stored in memory

☒ Writing all applications to use direct memory access to bypass the cache, rather than allowing the automatic CPU logic to control memory use

☐ Storing data you'll be working with in order in one-dimensional arrays

☐ When working with large multidimensional arrays, avoiding the use of leading dimensions whose size is a power of two

Nice work

Correct; This approach is less efficient because reading data directly from main memory is much slower for nearly all numerical applications.
3. What is the best guideline to improving program efficiency using array tiling?

1 / 1 point

☐ Decrease tile size so less memory needs to be loaded all at once

☒ Using the command line or available documentation for the system the program will run on, find out the details about each cache directory, then make tiles the same size as the largest cache that is specific to a single core

☐ Increase tile size as much as possible to break the data into a number of pieces that is a power of two

☐ Using the command line or available documentation for the system the program will run on, find out the details about each cache directory, then make tiles the same size as the largest cache that is shared across multiple cores

Nice work

Correct. Usually, the best place to start is by matching tile size with the largest cache used by a single core.
4. What is the ideal Bc ratio, considering the limitations of the Von Neumann architecture?

1 / 1 point

☒ Low data work and high arithmetic work, producing a low Bc ratio

☐ High data work and low arithmetic work, producing a low Bc ratio

☐ High data work and high arithmetic work, producing a balanced Bc ratio

☐ Bc is just a measurement tool - a Von Neumann computer can deal equally well with all Bc ratios

Nice work

Correct - most computers can perform arithmetic operations much faster than data operations, which makes a low Bc ratio optimal.
5. What is the purpose of cache memory in a computer's memory hierarchy?

1 / 1 point

☒ To hold data that are about to be used or have just been used by the CPU

☐ To act as a backup in case of power failure

☐ To replace main memory and serve as the primary storage

☐ To hold all of the program instructions and data

Nice work

Correct. Cache memory acts as a staging area for data before it reaches the CPU
6. Which of the following is true about cache lines?

1 / 1 point

☐ Cache lines are individual data items that are loaded into cache one by one

☐ Cache lines are dynamically allocated based on the size of the data being accessed

☒ Cache lines are groups of data that are contiguous in main memory, loaded together to take advantage of spatial locality

☐ Cache lines are fixed-size blocks that contain both data and instructions

Nice work

Correct. Cache lines are contiguous blocks of data loaded together to exploit spatial locality.
7. What is the purpose of prefetching in cache memory?

1 / 1 point

☐ To compress data in cache to increase the effective cache size

☐ To evict cache lines that are no longer needed to make room for new data

☐ To rearrange the order of cache lines for more efficient access

☒ To load additional cache lines before they are expected to be needed, reducing cache misses

Nice work

Correct. Prefetching loads extra cache lines in advance to reduce cache misses.)
8. Which of the following programming techniques can improve spatial locality for multidimensional arrays?

1 / 1 point

☒ Iterating over the array in row-major order for C/C++ or column-major order for Fortran

☐ Iterating over the array in column-major order for C/C++ or row-major order for Fortran

☐ Avoiding the use of multidimensional arrays altogether

☐ Using a random access pattern to access array elements

Nice work

Correct. This access pattern matches the memory layout, improving spatial locality
9. What is cache thrashing, and how can it be mitigated?

1 / 1 point

☐ Cache thrashing is when the cache becomes full and has to evict all lines; it can be mitigated by increasing the cache size

☐ Cache thrashing is when too many cache lines are needed at once, leading to repeated cache misses and evictions; it can be mitigated by avoiding arrays with leading dimensions that are powers of two

☒ Cache thrashing is when data is accessed in a random pattern, leading to poor spatial locality; it can be mitigated by using tiling techniques

☐ Cache thrashing is when the cache fails to load data correctly due to hardware errors; it can be mitigated by rebooting the system

Nice work

Correct. Cache thrashing is caused by specific access patterns, and can be mitigated by avoiding certain array dimensions.
10. What is the significance of the Bc ratio in evaluating algorithm performance?

1 / 1 point

☐ The Bc ratio represents the balance between cache hits and cache misses; a high Bc ratio (more cache hits) is desirable for efficient performance

☒ The Bc ratio represents the balance between arithmetic work and data work; a low Bc ratio (low data work, high arithmetic work) is desirable for efficient performance

☐ The Bc ratio represents the balance between main memory accesses and cache accesses; a high Bc ratio (more cache accesses) is desirable for efficient performance

☐ The Bc ratio represents the balance between single-core and multi-core performance; a low Bc ratio is desirable for efficient multi-core scaling

Nice work

Correct. A low Bc ratio indicates more arithmetic work relative to data work, which is efficient for modern computers.